

## **Italy**

### **Italian Competition Authority**

#### **I. Introduction**

The spread of digital technologies and innovation have profoundly transformed directional, managerial and operational processes in both businesses and public institutions, including Competition Authorities (CA).

Indeed, there are two main aspects to be considered for competition authorities, i.e., i) the collection and use of big data and artificial intelligence (AI) conditions the functioning of markets and has given rise to new ones, ii) AI offers great opportunities for improving the efficiency and effectiveness of the performed activities.

Therefore, CA must update their competencies by developing the capacity to analyze the conduct of firms, understand the functioning of markets, and identify possible competitive problems in an economy that is increasingly based on data and artificial intelligence, e.g. understand data management procedures of firms, analyze algorithms, examine the evolutions in the functioning of markets.

Additionally, CA can improve the efficiency and effectiveness of their operational performance with AI and Large Language Models (LLMs).

Accordingly, European competition authorities are adapting to technological developments by establishing offices, units and working groups dedicated to data science and AI.

#### **II. The Italian Data Science Unit (I-DSU)**

The Data Science Unit of the Italian Competition Authority (ICA) was recently established in the Directorate of the Chief Economist with the goal of keeping up with the challenges of the digital age. The Unit is responsible for supporting all institutional activities through expert advice and for the development of specific projects on data, computer techniques, artificial intelligence and algorithms. The Unit collaborates on investigative activities of cases, taking care of in-depth computer analyses and implementation of innovative methodologies for analyzing markets with advanced data analysis and artificial intelligence techniques.

The I-DSU is coordinated by the Chief Economist and is composed by one scientific officer / senior data scientist, two new data scientists that recently joined the Italian Competition Authority, and one fixed term legal expert. The members of the Data Science Unit have backgrounds in engineering, astrophysics, statistics, data management, AI, and cloud contracts. They have PhDs and have worked in international environments both in academia and industry. Furthermore, Data Science skills can be found also in other directorates within the Authority, as the latter has constantly promoted training on the subject.

The computational resources of the I-DSU are and will be the ones provided by the Italian Cloud for the Public Administration (PSN), a national and secure environment that is able to warrant confidentiality and data protection.

The I-DSU has five main lines of action:

- Providing internal consulting to investigative directorates in cases where technology plays a relevant role;
- The development of data acquisition and analysis techniques to support investigative proceedings and market analysis;
- The analysis of the use of data and algorithms in firms and the implications for competition and consumers;
- The development of projects for the use of artificial intelligence to support the institutional activities;
- Managing international relations with similar working groups established in other competition authorities and any collaborative project with research institutions and other national organizations.

The reader may refer to Figure I for a short overview of the main activities of the I-DSU.

### **III. The activities of the I-DSU**

The activities in the previously specified line of actions require an intensive use of AI and data science techniques. In particular, the I-DSU is actively working on the development of methods for data collection, management and analysis of digital markets and algorithms via web scraping techniques.

Indeed, the I-DSU has provided advice to investigative directorates by participating in the Big Data Market Study, and in the sector inquiry about the use of pricing algorithms in passenger air transportation. Such investigation required the

usage of advanced web-scraping techniques to collect huge amounts of data on flight prices and extra services, to be subsequently analyzed.

Moreover, it has participated in several cases where technology played a major role, e.g., the analysis of booking and pricing algorithms of the Italian railways, the reverse engineering of ranking algorithms by hotel and e-commerce platforms, and data portability.

Additionally, the I-DSU supports institutional activities with the development, and maintenance of the online request for information platform (e-rfi), with advanced data analysis techniques by high level programming languages (e.g., Python), with leveraging Geographic Information Systems (GIS) for mergers analysis, and algorithmic auditing services.

The I-DSU designed and developed an online platform, to obtain information from companies by means of secure online forms. It is a web-based platform to perform online requests for information from companies in order to acquire and process data within the framework of a proceeding. The platform works with a confidential web page and a flow of certified electronic mail messages to collect the information requested from a company, record them and assign them to a case. The platform has a simple and intuitive graphic interface and it also has inherent reporting and analysis capabilities. Thanks to a data schema, to the structured format and to the relational database, the collected data can be analyzed not only by the platform itself, but also by statistical and office automation tools. Thanks to a reporting system via certified e-mail, the web platform is able to send to the companies a report of the answers provided, which is also used as proof of receipt of the data, and to automatically send these answers to the Authority's registry. The request for information system was developed in-house by adopting an open source framework for online survey. The system runs on a cloud computing space of the authority provided by the Italian Cloud for the Public Administration (PSN).

Among the applications in this area of activity, the DSU is currently applying GIS to merger analysis. A tool based on Google Maps API is currently under development to enable officers to estimate distances and route times between multiple geographical points.

Another important field of activity is the monitoring and screening via AI and data science techniques. The I-DSU is developing tools to monitor and screen possible restrictive conduct of competition, i.e., automated screening of possible bid-rigging attempts, and non-notified mergers screening.

Particular focus has the bid-rigging screening, as the I-DSU is actively contributing to the activities carried out by a dedicated working group for analyzing tender data by means of multiple red flags, to identify possible bid-rigging phenomena.

Moreover, the I-DSU set up a system by using closed source databases and business intelligence platforms for monitoring mergers that are below the notification threshold.

Additionally, the I-DSU is contributing to the e-discovery process by the analysis of documents collected during dawn raids and will support the Directorate for Information Resources and Inspections in the acquisition of state-of-the-art software and in the integration of AI tools.

Furthermore, the I-DSU plans to participate in the selection, the design and the development of AI tools to increase the efficiency of institutional activities. Indeed, data science and artificial intelligence tools are able to support day-to-day activities of officers. A major project is to deploy an advanced text assistant for document analysis based on Large Language Models (LLMs) that is able to provide summarization, information extraction, named entity recognition, classification comparison, and text generation and that leverages AI for automatic natural language processing. Open-source models and commercial solutions be taken into consideration. Other projects concern the semantic search in order to facilitate the information retrieval and leveraging speech to text AI methods for automatic transcription of internal meetings and hearings. The selection of AI systems is restricted to the ones compatible with the data confidentiality requirements and to the Italian AI act guidelines.

Finally, the I-DSU fosters international relations and collaborations in the fields of data science and AI through participation in working groups and European projects.

It participates actively in the European Competition Network (ECN) Working Group about Digital Investigation and Artificial Intelligence and the International Competition Network (ICN) Technologist Forum. To conclude, it is engaged in a Technical Support Instrument (TSI) European project entitled ‘Supporting Digital Transformation in Competition Law Enforcement with Artificial Intelligence Methods and Technologies,’ where computational methods for competition analysis and enforcement will be further deepened and assessed.



**Figure I:** The main activities of the Italian Data Science Unit